

The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

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Abstract

Kolmogorov's probability fixes the event algebra to be Boolean, so every event is decidable. This is a poor fit for the semidecidable properties, termination, reachability, that verification cares about. Following the Cox–Jaynes interpretation of probability as extended logic, we define a construct for extending a logic (treated as a parameter in our approach) to a calculus of probability, and specify criteria for when this works: (1) when the resulting calculus can reconstruct the original logic in the limiting case of certain (0– or 1– probability) events, and (2) when the construction, evaluated at classical logic, agrees with classical probability theory. Instantiating the construction at intuitionistic logic gives valuations on frames, complete Heyting algebras of open sets, the algebras of verifiable propositions. In this setting, we define slack, $1 - v(A) - v(\neg A)$, the classical complement rule's deficit and boundary mass in the measure model. Slack pinpoints exactly when Dempster and Bayesian conditioning disagree. A separate obstruction, a five-element counterexample, is why Cox's sum rule cannot be derived and must instead be posited. We show that applying classical probability to semidecidable properties is equivalent to solving the halting problem. The whole development is machine-checked in Lean 4 over mathlib, in roughly 5,000 lines with no sorry.

Keywords: intuitionistic logic, probability, Cox's theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Kolmogorov's probability [Kolmogorov 1933] assigns numerical values representing uncertainty to events. The events to which the probability is assigned form a Boolean σ -algebra, therefore satisfying the law of excluded middle. The axioms these assignments satisfy make up the arithmetic of chance (non-negativity, normalization, σ -additivity).

This framework commits to logical axioms that computer science cannot commit to. Many properties of programs (does this computation halt, does this process reach a given state) are only *semidecidable*: a procedure confirms the property when it holds, and no procedure, in general, confirms its failure. The natural event algebra for a semidecidable property is a *frame*, the algebra of open sets of a space, and equivalently the algebraic semantics of intuitionistic logic. We show that applying Kolmogorov's probability (with its

underlying Boolean event algebra) to a semidecidable property is equivalent to solving the halting problem. This result is the pointwise counterpart of Chaitin's Ω [Chaitin 1975] example, which we interpret in the framework we develop here. In general, reasoning about the chance that a program has some property therefore needs an event logic that is not Boolean. We develop a mechanized framework that allows us to study probability calculi without committing to a fixed (classical) logic, and thereby select the probability calculus aligned with the logic of the objects it reasons about.

Probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003]. Different logics can be extended through the same construction. First, fix an event algebra, i.e. the chosen event logic in algebraic form. This algebra consists of the underlying set of events, together with the operations interpreting the logic's connectives. For the logics compared in this paper these are conjunction, disjunction, and negation, with the law of excluded middle holding in the classical case. Then let *valuations* be the maps from events to numbers that satisfy the axioms of monotonicity, normalization, and modularity. We specify the two sufficient and necessary conditions for when the extension construct we propose defines a probability calculus (i.e. degrees of certainty rather than an arbitrary grading of events). The first (1) is that the valuations taking only the values 0 and 1, the two degrees of complete certainty, reconstruct the underlying event logic (this is what we take "extended logic" to mean). The second (2) requires that the construction, evaluated at classical logic, returns the classical theory (this ensures that the new construction is a "generalization of classical probability"). This construction treats the logic of events as a *parameter*: the choice of event algebra determines an associated probability calculus.

Kolmogorov takes the event algebra to be Boolean, and his calculus is the value of the construction at Boolean logic. There, the law of excluded middle in the underlying logic makes it possible to show that every valuation obeys the *complement rule* $P(A^c) = 1 - P(A)$. A motivating example for varying the logic parameter is quantum uncertainty calculus, whose event algebras are the orthomodular lattices [Birkhoff and von Neumann 1936; Gleason 1957]. Here, we study the construction extending intuitionistic logic, whose event algebras are frames: complete Heyting algebras, the lattices of open sets. (1) The all-or-nothing valuations are exactly the two-valued models of the logic (Section 3). (2) Complete Boolean algebras are exactly the frames satisfying excluded middle, so the classical theory embeds as a special case.

Given a valuation $v : \Omega \rightarrow [0, \infty]$ on a frame Ω , for all a, b , by modularity, $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$. The pseudocomplement $\neg a$ satisfies $a \sqcap \neg a = \perp$ but not, in general, $a \sqcup \neg a = \top$, so the complement rule degrades to $v(a) + v(\neg a) \leq 1$, with a gap

$$\text{slack}(a) = 1 - v(a) - v(\neg a).$$

Classically, this is the width of the Dempster–Shafer belief/plausibility interval [Dempster 1967; Shafer 1976], quantifying ignorance. In our system it instead quantifies the mass of the part of a that is neither confirmed nor refuted, the probabilistic image of the semidecidable gap above. On the opens of a space with measure μ , the slack of an open U is $\mu(\partial U)$, the mass of its topological boundary. For verification this matters concretely: knowing how likely a property is to hold no longer tells you how likely it is to fail, and the gap between the two, the slack, is now also required to answer that question.

How the change of logic affects statistical reasoning can be stated operation by operation. The conditional $v(a \sqcap b)/v(b)$, the product rule, and Bayes symmetry hold verbatim. The classical denominator of Bayes' theorem, $P(B) = P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over partitions. The complement trick $P(A) = 1 - P(\neg A)$ degrades to the interval $[v(A), 1 - v(\neg A)]$ of width $\text{slack}(A)$. Dempster's rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes. The value of an event A itself is read as $P(\text{int } A)$, so a nonatomic distribution queried on intervals still behaves classically: an interval's boundary is two points, carrying no mass. The two readings differ only at boundaries carrying mass and at semidecidable hypotheses.

Cox's original theorem required repair after Halpern's counterexample [Cox 1946; Halpern 1999; Horn 2003]. Even with this repair, the direct translation of Cox's theorem to intuitionistic logic is *unsatisfiable*. This is because the sum rule is irreducible: a disjunction's value is provably not a function of its disjuncts' values, so modularity must be posited rather than derived. Weak excluded middle, an assumption strictly between intuitionistic and classical logic, does not restore the reducibility of the sum rule: the counterexample frame itself satisfies weak excluded middle.

Contributions.

1. **Weak excluded middle** (Sections 2 and 7). Global weak excluded middle on the opens of X is exactly extremal disconnectedness of X , for compact Hausdorff X exactly the Stonean spaces.
2. **Two characterizations** (Sections 3 and 5). The complement rule holds for *every* valuation iff the frame satisfies excluded middle, the exact content of Van Horn's classical negation axiom R3 [Horn 2003]. Dempster

and Bayesian conditioning agree iff the evidence has zero slack.

3. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core it is a belief function.
4. **Cox's theorem, dissected** (Section 6, Appendix C). The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half, a theorem classically, admits no derivation here (a five-element counterexample) and must be posited.
5. **Two grounding models** (Section 7). Any classical measure, read on the opens, is a valuation whose slack is boundary mass. A slack-free readout of halting is uncomputable, and on Cantor space the value of the halting event is Chaitin's Ω . Slack is submodular yet does not determine the valuation.
6. **The monad dichotomy** (Section 8). Product valuations are unique on $\text{Low}(\mathbb{N} \times \mathbb{N})$ and provably non-unique on $\text{Set}(\mathbb{N} \times \mathbb{N})$, where Fubini fails. The countably-presented fragment composes associatively and commutatively.
7. **A first formal treatment**. Everything above is machine-checked in Lean 4 over mathlib, to our knowledge the first mechanization of non-additive probability over intuitionistic logic.

Mechanization. The development is carried out in Lean 4 [de Moura and Ullrich 2021] over mathlib [mathlib Community 2020], in roughly 5,000 lines with no sorry, and is included in full as anonymous supplementary material.¹ Two points are specific to a proof assistant. We separate *object-level* classicality, excluded middle in the frame Ω , from the *meta-level* classicality mathlib inherits through `Classical.choice`, track which results use which, and equip the one essential non-Boolean model with a computable decidability instance. The product-rule half of Cox's theorem needs an Aczél/Hölder regratuation absent from mathlib, which we build from scratch.

2 Valuations on Frames

2.1 The Event Logic

Definition 1 (frame). A *frame* is a complete lattice Ω in which finite meets distribute over arbitrary joins,

$$a \sqcap \left(\bigsqcup_{i \in I} b_i \right) = \bigsqcup_{i \in I} (a \sqcap b_i)$$

for every $a \in \Omega$ and every family $\{b_i\}_{i \in I} \subseteq \Omega$. Equivalently, it is a complete Heyting algebra. By $x \Rightarrow y$ we denote the Heyting implication in a frame, and $\neg a := a \Rightarrow \perp$ is the pseudocomplement of $a \in \Omega$. Also, a^c denotes a complement

¹Claude (Anthropic) was used to assist with drafting and debugging the Lean formalization.

of a , if it exists, i.e. an element such that $a \sqcap a^c = \perp$ and $a \sqcup a^c = \top$.

Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale* for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudocomplement $\neg a$ of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap \neg a = \perp$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup \neg a = \top$ is available only in Boolean algebras. An element with $a \sqcup \neg a = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The Valuation Structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 2 (valuation). A *valuation* on a frame Ω is a *monotone* function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$ (*normalization*), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
    = toFun (a ⊔ b) + toFun (a ⊓ b)
```

This axiom set is not ours: Weatherston’s axioms for intuitionistic probability are exactly these [Weatherston 2003], and the valuations of constructive measure theory [Coquand and Spitters 2009; Vickers 2008] are their Scott-continuous instances (Section 9).

Values live in the extended non-negative reals $[0, \infty]$, and normalization confines them to $[0, 1]^{\text{Lean}}$. The conventions this choice carries, unconditional summation and truncated subtraction, are an artifact of the mechanization and are recorded in Appendix A.

2.3 Slack

Modularity applied to the disjoint pair $(a, \neg a)$ collapses to $v(a) + v(\neg a) = v(a \sqcup \neg a)^{\text{Lean}}$. So, the join need not be $\top$:

Theorem 3 (the complement bound). *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(\neg a) \leq 1$.*

```
theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v a^c ≤ 1
```

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\text{slack}(a) = 1 - (v(a) + v(\neg a)) = 1 - v(a \sqcup \neg a)$$

is the valuation of the undecided region. It decomposes canonically. The pair $(\neg\neg a, \neg a)$, double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($\neg a \sqcup \neg\neg a = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Definition 4 (regular element). An element $a \in \Omega$ is *regular* (or $\neg\neg$ -stable) if $a = \neg\neg a$.

Theorem 5 (slack decomposition^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{slack}(a) = \text{dnGap}(a) + \text{deMorganGap}(a)$, where $\text{dnGap}(a) = v(\neg\neg a) - v(a)$ is 0 if a is $\neg\neg$ -stable (regular), and $\text{deMorganGap}(a) = 1 - v(\neg\neg a \sqcup \neg a)$ is 0 if weak excluded middle holds at a .*

The single Dempster–Shafer “ignorance” number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). Constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel’s raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most $\neg\neg a$, not a itself, and dnGap is exactly the gap between confirming $\neg\neg a$ and confirming a itself.

Because dnGap and deMorganGap are themselves defined from v , the apparent three unknowns $v(a)$, $v(\neg a)$, and $v(\neg\neg a)$ are not independent: they are part of a three-constraint system, yielding the following result:

Theorem 6 (slack in closed form^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$,*

$$\text{slack}(a) = (v(\neg\neg a) - v(a)) + (1 - v(\neg\neg a) - v(\neg a)),$$

where $v(a) \leq v(\neg\neg a)$.

The inequality follows from monotonicity applied to $a \leq \neg\neg a$, a fact of every Heyting algebra, and shows the first summand is exactly the difference, not the floor produced by $[0, \infty]$ ’s truncated subtraction. The split into a regularity defect and a De Morgan defect follows from monotonicity and modularity alone, with no further axiom about ignorance required. This contrasts with the usual Dempster–Shafer picture, where the mass assigned to non-singleton, ambiguous sets is an extra primitive.

Theorem 6 expresses slack pointwise in terms of v . The converse, whether slack in turn determines v , is more delicate. One constraint is automatic:

Theorem 7 (slack is submodular^{Lean}). *For every valuation v on a frame Ω and every $a, b \in \Omega$,*

$$\text{slack}(a \sqcup b) + \text{slack}(a \sqcap b) \leq \text{slack}(a) + \text{slack}(b).$$

The proof runs v 's own modularity twice, once at (a, b) and once at $(\neg a, \neg b)$, and combines the two results with the De Morgan laws: the equality $\neg(a \sqcup b) = \neg a \sqcap \neg b$, which holds in every Heyting algebra, and the inequality $\neg a \sqcup \neg b \leq \neg(a \sqcap b)$, monotone under v but not in general an equality. Any function proposed as the slack of some valuation must therefore be submodular.

Remark 8 (non-recovery^{Lean}). Submodularity is necessary, not sufficient: Section 7.2 exhibits two valuations on the same frame with identical slack functions.

Remark 9 (slack under weak excluded middle^{Lean}). The two gaps, dnGap and deMorganGap , computed at a single element, are logically independent. However, if weak excluded middle is imposed on the frame for every a ($\neg\neg a \sqcup \neg a = \top^{\text{Lean}}$), then deMorganGap vanishes identically and $\text{slack}(a) = \text{dnGap}(a)$ for every valuation and every a . This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish^{Lean}. A concrete instance appears in Section 7.

2.4 What Survives without Excluded Middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \perp$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b) / v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b) v(b) = v(b \mid a) v(a)$ hold^{Lean}. *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \perp$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, \neg a\}$. That family need not join to $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack^{Lean}.

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$ -stable (*regular*) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $\neg\neg(a \sqcup b)$, for which only the inequality $v(a) + v(b) \leq v(\neg\neg(a \sqcup b)) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^c) = 1$ holds

exactly^{Lean}. On a connected locale that fragment can be as small as $\{\perp, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and Kolmogorov is recovered^{Lean}.

Mixtures. Valuations are closed under finite convex combination^{Lean}, with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Classicality in Theory and Metatheory

Classicality enters this work twice: in the metatheory, the logic of the proof assistant itself, and in the object theory, the frame's own logic.

Lean's meta-theory. Lean's foundations include `Classical.choice`, and `mathlib` is built on it throughout. For example, the order relation on $\mathbb{R}$ and $[0, \infty]$ has no computable Decidable instance in general, so any library lemma that case-splits on $x \leq y$ for reals resolves that split via `Classical.propDecidable`. A formalization that never invokes excluded middle at the level of frame elements $a \in \Omega$ can therefore still be classical through the arithmetic it borrows from the library. We track this separately from object-level classicality throughout the paper. Theorem 20 of Section 4 rests only on this ambient baseline and contains no explicit case split. The Scott-continuity recovery theorem (Theorem 19) invokes Lean's `classical` tactic directly, and the explicit Cox model of Section 6 goes the other way, replacing the default with an explicit computable Decidable instance to keep that one instance genuinely non-classical.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A . Consistency forces the involution $S \circ S = \text{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation is double-negation elimination, so *R3* is where classical logic enters every known Cox-style derivation. To do this, we first isolate the exact content of *R3*'s conclusion as a predicate on valuations, then prove equivalence to excluded middle.

That predicate is *classical negation*: a valuation v has classical negation if $v(a) + v(\neg a) = 1$ for every $a \in \Omega$.

Theorem 10 (the *R3* characterization^{Lean}). *For every frame Ω , every valuation on Ω has classical negation iff $a \sqcup \neg a = \top$ for every $a \in \Omega$.*

The forward direction is a calculation. The content is the converse: wherever excluded middle fails, a valuation with genuine slack must be *constructed*. Suppose $a \sqcup \neg a \neq \top$.

Definition 11 (ideal, prime ideal). An ideal $J \subseteq \Omega$ is downward closed and closed under finite joins. J is *prime* if $\top \notin J$ and $a \sqcap b \in J$ implies $a \in J$ or $b \in J$.

The prime ideal separation theorem (mathlib's `DistribLattice.prime_ideal_of_disjoint_filter_ideal`) yields a prime ideal J containing $a \sqcup \neg a$ but not $\top$. The indicator of J 's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(\neg a) = 0 \neq 1$.

Sharp valuations are the points of the locale. The construction above is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$ -valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 12 (sharp valuations are points^{Lean}). For every valuation v on a frame Ω , v is sharp iff v is the complement-indicator of a prime ideal.

The correspondence is given by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 10's construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of Theorem 12 sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008]:

Theorem 13 (Scott-continuous sharp valuations^{Lean}). For every valuation v on a frame Ω , v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.

The characterization and the two correspondences together locate the object-level classicality precisely: whenever excluded middle fails at some a , the failure occurs at a point of the locale.

4 Valuation Representation

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott continuity marks the boundary between where it still holds and where it does not.

4.1 Finite Frames Representation

Theorem 14 (Birkhoff duality [Birkhoff 1937]). Every finite distributive lattice is isomorphic to the lattice of lower sets of a finite poset: its sup-irreducible elements form a poset P whose

lower sets reconstruct the lattice, and every finite poset arises this way from its own lattice of lower sets.

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (`OrderIso.lowerSetSupIrred`), every finite frame is the lattice of lower sets of P . We fix the following notation, used throughout the rest of the paper. A *lower set* $L \subseteq P$ satisfies $p \in L, q \leq p \Rightarrow q \in L$, and an *upper set* is defined dually. Lower sets are always denoted L , and upper sets U (opens of a topological space are still U, V, W , as in traditional usage). The *principal lower set* at p is $\downarrow p = \{q : q \leq p\}$, the *strict principal lower set* at p is $\Downarrow p = \{q : q < p\}$, and the *principal upper set* at p is $\uparrow p = \{q : q \geq p\}$. We prove the finite representation theorem below on the lower sets of P . Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets^{Lean}. Let us define the *mass* of a point as the discrete derivative of v^{Lean} :

$$\text{mass}(p) = v(\downarrow p) - v(\Downarrow p).$$

The masses recover v and sum to 1:

Theorem 15 (finite representation^{Lean}). For a finite P , any valuation v on $\text{Low}(P)$, and any $L \in \text{Low}(P)$, $v(L) = \sum_{p \in L} \text{mass}(p)$, and $\sum_p \text{mass}(p) = 1$.

The proof works by induction on $|L|$: repeatedly remove a maximal point p from L , and use modularity applied to $(L \setminus \uparrow p, \downarrow p)$ to show each removal changes v by exactly $\text{mass}(p)$. The mass function is a classical probability mass function on P^{Lean} . And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \text{int}$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7. The intuitionistic mass also agrees with the classical notion of the mass of a point: reading $v = \mu \circ \text{int}$,

$$\text{mass}(p) = \mu(\downarrow p) - \mu(\Downarrow p) = \mu(\{p\}),$$

since $\downarrow p$ is the disjoint union of $\Downarrow p$ and $\{p\}$ and μ is defined on all subsets of P .

Theorem 15 represented v as a sum of point masses. The dual view represents v as a mixture of sharp *point-valuations*. For each $p \in P$, let δ_p be the valuation with $\delta_p(L) = 1$ if $p \in L$, and $\delta_p(L) = 0$ otherwise:

Theorem 16 (mixture characterization^{Lean}). On a finite frame, every valuation v satisfies $v = \sum_p \text{mass}(p) \cdot \delta_p$.

Together with the fact that any mixture of valuations is again a valuation^{Lean}, this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 Infinite Boundary Representation via Scott Continuity

Both halves of the finite theorem fail for infinite frames:

Example 17 (obstruction^{Lean}). On $\text{Low}(\mathbb{N})$, the indicator v of $\top$ is a valuation with $\text{mass}(p) = 0$ for every point p , and $v(\top) = 1$.

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\downarrow n)$ over the finite stages. In order to recover the representation, we introduce Scott continuity:

Definition 18 (directed set, Scott continuity). A subset $D \subseteq \Omega$ is *directed* if, for any finite subset $F \subseteq D$, there exists an element $d \in D$ that is an upper bound of F (taking $F = \emptyset$ shows a directed set is nonempty). A valuation v is *Scott-continuous* if $v(\bigsqcup D) = \sup_{a \in D} v(a)$ for every directed D .

The $\top$ -indicator of Example 17 is not Scott-continuous^{Lean}, and Scott continuity is exactly what restores the representation:

Theorem 19 (recovery^{Lean}). Let v be a valuation on $\text{Low}(\mathbb{N})$. If $v(\top) = \sup_n v(\downarrow n)$ then $v(\top) = \sum_n \text{mass}(n)$.

In general, on any poset all of whose principal lower sets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum_p \text{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum_p \text{mass}(p)$ would still evaluate to 0. Theorem 20 rules out $\sum_p \text{mass}(p) > v(\top)$ unconditionally, so the subtraction below is never truncated:

Theorem 20 (general decomposition^{Lean}). For every valuation v on $\text{Low}(P)$ and an arbitrary poset P , $\sum_p \text{mass}(p) \leq v(\top)$.

By Theorem 20 the remainder $v(\top) - \sum_p \text{mass}(p)$ is a genuine difference. Every valuation thus splits unconditionally into an *atomic* part carried by points ($\sum_p \text{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum_p \text{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$ -indicator is the purely diffuse one. The proof of Theorem 20 re-runs the maximal-point induction on each finite subset of P , bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum_p \text{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same one as behind the undecidability of Section 7.

5 The Dempster–Shafer Reading

Sections 2–4 independently arrive at the same structure as Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976],

including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it locates the precise point where excluded middle makes a difference.

5.1 Total Monotonicity via Constructive Inclusion–Exclusion

The $\neg\neg$ -stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $\neg\neg(a \sqcup b)$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set.

Definition 21 (belief function, plausibility function). Let B be a Boolean algebra. A DS *belief function* on B is a monotone map $\text{Bel} : B \rightarrow [0, 1]$ with $\text{Bel}(\perp) = 0$ and $\text{Bel}(\top) = 1$ that is *n-monotone* for every $n \geq 2$: for all $A_1, \dots, A_n \in B$,

$$\text{Bel}\left(\bigvee_{i=1}^n A_i\right) \geq \sum_{\emptyset \neq T \subseteq \{1, \dots, n\}} (-1)^{|T|+1} \text{Bel}\left(\bigwedge_{i \in T} A_i\right).$$

Its *plausibility function* is the map $\text{Pl} : B \rightarrow [0, 1]$ given by $\text{Pl}(A) = 1 - \text{Bel}(\neg A)$.

We show v , restricted to its Booleanization, is such a belief function. The 2-monotone case is immediate from modularity^{Lean}. The full tower follows from the following result:

Theorem 22 (inclusion–exclusion^{Lean}). For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:

$$v\left(\bigsqcup_{i \in s} a_i\right) = \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\bigsqcup_i a_i$ on the equality's other side with its double negation $\neg\neg(\bigsqcup_i a_i)$, the Booleanization's own join, yields the following inequality:

Theorem 23 (total monotonicity^{Lean}). For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$:

$$\sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right) \leq v\left(\neg\neg\left(\bigsqcup_{i \in s} a_i\right)\right).$$

The Booleanization's join is given by $\neg\neg \circ \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 23 states exactly the ∞ -monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 23 from Theorem 22 follows from the monotonicity along $a \leq \neg\neg a$. Here, inclusion–exclusion does *not* fail constructively. It is exact for the intuitionistic disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $\neg\neg$ to the join on the

equality's right-hand side can only enlarge it. This follows from the fact that $a \leq \neg\neg a$ and v are monotone, which turns the equality into the inequality of Theorem 23. The amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Definition 21 is the classical notion, stated on an abstract Boolean algebra. In our setting each of its ingredients is already present: the Boolean algebra is the Booleanization, the belief function is v itself, restricted to the regular elements and evaluated against the Boolean join $\neg\neg \circ \sqcup$, and the plausibility function is $\text{Pl}(a) = 1 - v(\neg a)$. Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the join used to state the tower. The belief–plausibility interval is governed by the slack:

Theorem 24 (interval width^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{Pl}(a) - v(a) = \text{slack}(a)$.*

5.2 The Conditioning Characterization

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it is the localic posterior $v(a \sqcap b)/v(b)$ of Section 2.

Definition 25 (Dempster's rule of conditioning^{Lean}). For a valuation v on a frame Ω , elements $a, b \in \Omega$, and evidence of nonzero plausibility, $v(b) \neq 0$, *Dempster's rule of conditioning* [Shafer 1976] states that

$$v(a \parallel b) = \frac{v(a \sqcup \neg b) - v(\neg b)}{1 - v(\neg b)}.$$

The hypothesis $v(b) \neq 0$ makes the denominator nonzero, since $v(b) + v(\neg b) \leq 1$ gives $v(b) \leq 1 - v(\neg b)$.

That is, the DS conditioning rule discards the mass incompatible with the evidence and renormalizes by its plausibility. Both the geometric posterior of Section 2 and Dempster's rule of Definition 25 are again valuations. Their modularity is the frame distributivity $(a \sqcup \neg b) \sqcap (a' \sqcup \neg b) = (a \sqcap a') \sqcup \neg b$. Their disagreement is given by the *excluded-middle gap* $\text{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup \neg b))$, i.e. the mass of a outside b 's instance of excluded middle:

Lemma 26 (the join–complement identity^{Lean}). *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup \neg b) = v(\neg b) + v(a \sqcap b) + \text{emGap}(a, b)$.*

Dempster's numerator therefore exceeds Bayes's by exactly $\text{emGap}(a, b)$. This gap depends on a , but whether it vanishes at every a depends only on the slack at the evidence b :

Theorem 27 (the conditioning characterization^{Lean}). *For every valuation v on every frame Ω and every $b \in \Omega$ with $v(b) \neq 0$, $(v(a \parallel b) = v(a \sqcap b)/v(b) \text{ for every } a \in \Omega) \text{ iff } \text{slack}(b) = 0$.*

The hypothesis $\text{slack}(b) = 0$ eliminates the gap $\text{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(\neg b) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(\neg b))$, which is 1 only if the slack vanishes. Theorem 27 is the valuation-relative analogue of Theorem 10: both are iff statements about the same proposition, $b \sqcup \neg b = \top$.

We call the frame-level fact *static* and the valuation-level fact *dynamic*: the former is an equation in the frame itself, between $b \sqcup \neg b$ and $\top$, with no valuation involved, while the latter tests the same proposition through v , since $\text{slack}(b) = 0$ is exactly $v(b \sqcup \neg b) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup \neg b$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence's boundary carries mass.

6 Cox's Theorem

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from conditions on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regreduation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 is excluded middle. Dropping R3 leaves the product rule unaffected, since it is logic-free, and requires the sum rule to become an axiom rather than a theorem. Separately, the naive constructive statement, demanding a regreduation strictly monotone on all of $\mathbb{R}$, is unsatisfiable (Section 6.3).

6.1 A Cox Structure without Negation

Definition 28 (Cox model^{Lean}). A *Cox model* on a frame Ω is a plausibility assignment $\text{pl} : \Omega \times \Omega \rightarrow \mathbb{R}$, written $\text{pl}(a, b)$ for the plausibility of a given b , together with a *conjunction functional* $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, such that $\text{pl}(\cdot, c)$ is monotone for every c , the boundary conditions $\text{pl}(\perp, c) = 0$ and $\text{pl}(\top, c) = 1$ hold for every c , the product rule $\text{pl}(a \sqcap b, c) = F(\text{pl}(a, b \sqcap c), \text{pl}(b, c))$ holds, F is associative, and $x \mapsto F(x, y)$ is strictly monotone for every $y > 0$.

The missing R3 axiom can be observed in the omission of any constraint on the relation between $\text{pl}(\neg a, b)$ and $\text{pl}(a, b)$. The definition restricts the monotonicity of F to $0 < y$ to avoid a contradiction with the fact that the boundary axioms force $F(x, 0) = 0$ for all x . An explicit model on the non-classical chain $[0, \infty]$ satisfies this constraint^{Lean}, as $a = \top$ is provably decidable.

6.2 The Sum Rule Is Irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general:

Theorem 29 (the sum rule is irreducible^{Lean}). *Let Ω be the five-element frame of lower sets of the poset $o < a, o < b$ with a, b incomparable. Then there is a monotone, normalized assignment q on Ω that is additive on disjoint joins yet not modular, and no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y . Moreover, the pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.*

Unlike for a conjunction, a disjunction's plausibility cannot generally be expressed as its disjuncts' plausibilities via conditioning. So, a constructive Cox theorem *must* posit the sum-rule half. Modularity, which says that the calculus is closed under mixing certainties, is what posits this, justified by Theorem 16. The failure is not repaired by strengthening the logic short of Boolean: the counterexample frame satisfies weak excluded middle^{Lean}, since every nonempty lower set contains the bottom point o , so the irreducibility persists under that intermediate assumption.

6.3 Cox Model Theorem

A regratuation is a map $g : \mathbb{R} \rightarrow [0, \infty]$ with $g(0) = 0$. No such map is strictly monotone on all of $\mathbb{R}$, since strictness would force $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the same domain restriction that Halpern's critique of the classical theorem turns on. The corrected statement is:

Theorem 30 (the constructive Cox theorem^{Lean}). *For every frame Ω , every Cox model on Ω whose unconditional plausibility is modular regratuates to a valuation: there exist $v : \text{Valuation } \Omega$ and g strictly monotone on $[0, 1]$ with $g(0) = 0, g(1) = 1$, and for all $a \in \Omega, v(a) = g(\text{pl}(a, \top))$.*

The regratuation for the sum-rule half is straightforward ($g = \text{ENNReal.ofReal}$). The product-rule half, showing F regratuates to multiplication, rests on Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω . `mathlib` has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity, so we build the analytic core from scratch, obtaining a normalized, strictly monotone multiplicative generator on the positive cone^{Lean}. The construction also weakens Hölder's classical hypotheses [Hölder 1901]: the Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility follows from the intermediate value theorem^{Lean} applied to a plain divisibility hypothesis^{Lean}. The constructive/classical divide is therefore *entirely* in the sum rule: nothing in the product-rule apparatus mentions Ω . Appendix C gives the construction. The residual analytic gap is discussed in Section 10.

7 Two Grounding Models

Two concrete models demonstrate that our abstract theory is non-degenerate. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the Interior Operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4: propositions are *open* sets, and the modal necessity operator $\Box$ of S4 is interpreted as the topological interior, written $\text{int}(\cdot)$ below [Gödel 1933; McKinsey and Tarski 1944]. Our valuations quantify that same translation:

Theorem 31 (measures as valuations^{Lean}). *Let μ be a probability measure on a space $X, A \subseteq X$, and define $v(A) = \mu(\text{int } A)$. Since $\text{int } U = U$ for open U , v restricts on the frame $\mathcal{O}(X)$ to $U \mapsto \mu(U)$, and this restriction is a valuation.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\text{int } U^c)$, which is the measure of the boundary ∂U . We can now prove that, in general, slack is exactly the boundary mass:

Theorem 32 (slack is boundary mass^{Lean}). *For any probability measure μ on a topological space X and any open U , $\text{slack}(U) = \mu(\partial U)$.*

The Heyting complement of U in $\mathcal{O}(X)$ is the interior of its set complement. So, the undecided region, the complement of $U \sqcup \neg U$, is exactly $\overline{U} \setminus U = \partial U$. The general sub-additivity inequality of Section 2 still holds here^{Lean}. This identification of slack with boundary mass also exhibits the modal identification $\text{Bel} = P(\Box \cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 15 is the converse in the finite case (in the general case, the converse is the open problem, see Section 4).

To characterize weak excluded middle spaces, we give the (standard) definitions required:

Definition 33 (extremally disconnected). A topological space X is *extremally disconnected* if the closure of every open set is again open.

Definition 34 (Stone space, Stonean space). The *Stone space* of a Boolean algebra B is the space of ultrafilters on B (equivalently, the 2-valued homomorphisms $B \rightarrow \{0, 1\}$), topologized by the clopen basis $\{U : b \in U\}$ for $b \in B$. *Stone duality* [Johnstone 1982] identifies Boolean algebras with exactly the compact, Hausdorff, totally disconnected spaces arising this way. A *Stonean space* is a compact Hausdorff space that is also extremally disconnected. Equivalently, it is the Stone space of a *complete* Boolean algebra.

Remark 35 (weak excluded middle on opens^{Lean}). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For

$\Omega = O(X)$, this global condition holds exactly when X is extremally disconnected.

An immediate corollary gives the characterization:

Corollary 36 (Stonean characterization^{Lean}). *Given a compact Hausdorff space X , weak excluded middle on $O(X)$ holds exactly when X is a Stonean space.*

7.2 Slack Does Not Determine the Valuation

Submodularity (Theorem 7) is a necessary condition on slack. It is not sufficient: distinct valuations, each satisfying every axiom a valuation must satisfy, can share a slack function.

Example 37 (boundaries in the three-point space^{Lean}). Let $\text{ThreePt} = \{p_A, p_B, p_Z\}$ carry the topology generated by the two singletons $\{p_A\}$ and $\{p_B\}$, so its opens are exactly $\emptyset$, $\{p_A\}$, $\{p_B\}$, $\{p_A, p_B\}$, and the whole space. For every open U , $\partial U \subseteq \{p_Z\}$.

This is a fact about the topology alone, independent of any measure: p_Z has no proper open neighbourhood, so it is the only point any open's frontier can contain.

Theorem 38 (non-recovery^{Lean}). *There exist valuations $v \neq v'$ on $O(\text{ThreePt})$ with $\text{slack}(v) = \text{slack}(v')$ as functions.*

For weights $a, b, c \geq 0$ with $a + b + c = 1$, the point measure $\mu_{a,b,c} = a \delta_{p_A} + b \delta_{p_B} + c \delta_{p_Z}$, read through Section 7.1's bridge, gives a valuation $v_{a,b,c}$ on $O(\text{ThreePt})$. By Example 37 and Theorem 32, its slack at an open U is $\mu_{a,b,c}(\partial U)$, equal to c when $p_Z \in \partial U$ and to 0 otherwise: a quantity depending only on c , not on how the remaining mass splits between a and b . So $v_{1/4,1/4,1/2}$ and $v_{1/8,3/8,1/2}$ have identical slack functions while disagreeing at $\{p_A\}$, where they take the values $1/4$ and $1/8$. Both are fully modular, monotone, and normalized: nothing in Theorem 7 or its proof distinguishes them.

Weak excluded middle does not rescue recoverability. If anything, the fully classical case is the worst one for it. When X is extremally disconnected, the De Morgan gap vanishes (Section 7.1) and slack degenerates to the double-negation gap alone, but on a space where excluded middle holds outright that gap vanishes too, so slack is identically 0 for every valuation. The zero slack function is then shared by every probability measure on X , which is the most extreme failure of recovery available.

7.3 The Computability Guard

The theory developed here is non-degenerate only if there are spaces whose slack does not vanish identically. Computability supplies them:

Definition 39 (Sierpiński topology, semidecidable proposition). The *Sierpiński space* $\mathbb{S}$ is the type Prop of propositions carrying the topology with exactly three opens: $\emptyset$, the set $\text{halts} = \{q \mid q\}$ of true propositions, and $\mathbb{S}$ itself [Escardó 2004; Rosolini 1986]. A proposition is *semidecidable* if it is an open in this topology.

The topology encodes verifiability. A point of $\mathbb{S}$ is a proposition, say “this computation halts.” The one nontrivial observation available is membership in *halts*, established by producing a proof (running the computation to termination). No open captures the complementary observation, and on the frame $O(\mathbb{S}) = \{\perp, \text{halts}, \top\}$ that Σ_1 asymmetry is a theorem:

Theorem 40 (the Σ_1 asymmetry^{Lean}). $\neg \text{halts} = \perp$.

Every nonempty open of $\mathbb{S}$ contains *True*, so $\perp$ is the only open disjoint from *halts*. Thus, no nonempty open of $\mathbb{S}$ is evidence that the computation does not halt.

We now instantiate the measure-to-valuation construction of Theorem 31 at $X = \mathbb{S}$, allowing us to compute the slack:

Example 41 (halting slack^{Lean}). Let $\mu = p \cdot \delta_{\text{True}} + (1-p) \cdot \delta_{\text{False}}$, and let v be the resulting valuation. Then

$$\text{slack}(\text{halts}) = 1 - p = \mu(\partial \{q \mid q\}) = \mu\{\text{False}\}.$$

Since *halts* is already open, $v(\text{halts}) = \mu(\text{halts}) = p$. The DS “ignorance” of Section 5 is here the mass of the single point *False*, the one outcome no finite observation can ever resolve. This probability is a fact about the model, not about anyone's ignorance.

Chaitin's Ω . The same bridge scales to algorithmic information theory. Fix a universal prefix-free machine U , *prefix-free* in that no halting program of U is a proper prefix of another, and *universal* in that it simulates every other prefix-free machine. Give 2^ω , the space of infinite bit strings, the fair-coin measure μ . The event H of having a halting program of U as a prefix is open, since halting is established by a finite prefix (Definition 39), and prefix-freeness makes H a disjoint union of cylinders, so $v(H) = \mu(H) = \sum_{p \text{ halts}} 2^{-|p|}$: exactly Chaitin's halting probability Ω [Chaitin 1975]. Its slack $\mu(\partial H)$ is the mass of strings approximated arbitrarily well by halting strings while never having a halting program as a prefix, and Ω 's classical uncomputability is the aggregate form of the pointwise obstruction above. The measure-theoretic core is mechanized on Cantor space with *mathlib*'s infinite product measure: basic cylinders are clopen with measure $2^{-|w| \text{Lean}}$, and a *finite* prefix-free set of words has pairwise disjoint cylinders of total measure $\sum_w 2^{-|w| \text{Lean}}$, the Kraft-inequality content underlying Ω . The universal machine itself, and the countable extension of the finite Kraft sum, are computability theory beyond our scope.

Definition 42 (halting problem). For a code c and input n , the *halting problem* is the predicate $H(c, n) =$ “the machine with code c halts on input n ,” (*ComputablePred*).

Slack cannot be eliminated by simply declaring a classical answer for each code. The family of valuations defined below attempts to do this:

Definition 43 (sharp classical readout). For a given n , the *sharp classical readout* is the family of slack-free valuations

$\rho_n(c) \in \text{Valuation}(O(\mathbb{S}))$, with $\rho_n(c)(\text{halts}) = 1$ if $H(c, n)$ holds, and $\rho_n(c)(\text{halts}) = 0$ otherwise.

However, this family is provably not computable:

Theorem 44 (the sharp readout is uncomputable^{Lean}). *For every n , the predicate $c \mapsto (\rho_n(c)(\text{halts}) = 1)$ is not computable.*

The proof identifies the predicate with $H(\cdot, n)$ and applies the undecidability of the halting problem (ComputablePred.halting_problem).

We can conclude that slack is not a defect of the constructive theory: any axiom strong enough to force classical collapse demands that its models compute the uncomputable. For the sum rule, this guard plays the role the explicit Cox-model instance plays for the product rule (Section 6): each axiom set in the paper comes with non-trivial models satisfying it.

8 Product Valuations and Kernel Composition

The structure through which classical probability supports sequential statistical reasoning is the Giry monad:

Definition 45 (Giry monad [Giry 1982]). *The Giry monad G sends a measurable space X to the space $G(X)$ of probability measures on X . Its unit sends a point $x \in X$ to the point mass at x , and its multiplication sends a measure M on $G(X)$ to the measure $A \mapsto \int m(A) dM(m)$.*

Probabilistic reasoning then composes in stages, and, when the monad is commutative, independent stages compose in either order. The question is whether the intuitionistic calculus supports the same structure.

A *frame homomorphism* is a map preserving finite meets and arbitrary joins. Pushforward along a frame homomorphism $f : \Omega \rightarrow \Omega'$ sends a valuation v on Ω' to $a \mapsto v(f(a))$ on Ω , and this preserves the valuation axioms, making $\text{Val}(-)$ a functor on locales, the localic counterpart of the functor part of the Giry monad [Giry 1982]. A frame homomorphism is the opposite of a locale map, so the functor laws are checked against identity and composition of frame homomorphisms^{Lean}. The point valuations δ_p (Section 4) provide the candidate (monad) unit. Theorem 15 shows that, on finite frames, Val agrees (on objects) with the classical finite-distribution monad on points. Flattening a valuation of valuations is integration, and integration in two variables requires product valuations. A *kernel* is a map assigning a valuation to each point of a locale, giving a conditional distribution. By Kock's identification of commutative monads with Fubini theorems [Kock 2012], kernels compose independently of order exactly when Val is a commutative monad.

A valuation u on a product frame is a *product* of valuations v and w when $u(A \times B) = v(A) \cdot w(B)$ for all rectangles $A \times B$. Let $\text{Low}(\mathbb{N} \times \mathbb{N})$ be the frame of lower sets of the grid $\mathbb{N} \times \mathbb{N}$, ordered component-wise. On this frame, products are unique:

Theorem 46 (product uniqueness^{Lean}). *Let v, w be valuations on $\text{Low}(\mathbb{N})$. Any two valuations on $\text{Low}(\mathbb{N} \times \mathbb{N})$ that are products of v and w are equal.*

By the inclusion-exclusion equality, two valuations agreeing on a $\sqcap$ -closed family agree on all finite joins of its members^{Lean}. The rectangles are closed under $\sqcap$, and in $\text{Low}(\mathbb{N} \times \mathbb{N})$ every lower set is a *finite* union of rectangles^{Lean}, so agreement on rectangles is agreement everywhere. Valuations carry no continuity axiom (Section 4), so the σ -additivity driving the classical uniqueness argument for product measures has no analogue here. The order structure of the frame does that work instead.

Theorem 47 (product non-uniqueness^{Lean}). *On the powerset frame $\text{Set}(\mathbb{N} \times \mathbb{N})$ there exist valuations v, w and two distinct products of v and w .*

An *ultrafilter* on $\mathbb{N}$ is a collection of subsets, closed under intersection and superset, containing exactly one of S and its complement for every $S \subseteq \mathbb{N}$. It is *free* when it contains no finite set. The map assigning 1 to the members of an ultrafilter and 0 to every other subset is a valuation^{Lean}. A rectangle $S \times T$ belongs to an ultrafilter on the product exactly when S and T belong to its images under the two coordinate projections^{Lean}. A free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ yields two ultrafilters on $\mathbb{N} \times \mathbb{N}$. A set belongs to the former when, for $\mathcal{U}$ -most first coordinates, its slice contains $\mathcal{U}$ -most second coordinates. A set belongs to the latter under the same test with the coordinates in the opposite order.

The two ultrafilters' valuations agree on *every* rectangle yet differ on the upper triangle $\{(m, m') \mid m < m'\}$ ^{Lean}. Integrating the same marginal data in the two orders thus disagrees. Fubini fails for valuations, as it does for non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock's correspondence, no canonical commutative multiplication exists for unrestricted valuations. This half of the dichotomy is not specific to intuitionism, since powerset-frame valuations are exactly the classical finitely additive probability measures [Rao and Rao 1983]. Dropping continuity is what admits the non-uniqueness. A monad therefore requires restricting either the frames or the valuations.

A first step in the second direction is to restrict Val to the valuations given as countable mixtures. For these, we can demonstrate that the kernels compose independently of order, the property that Theorem 47 refutes for unrestricted valuations:

Proposition 48 (countable kernel composition). 1. *A countable mixture of valuations is a valuation^{Lean}.*
2. *A mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman-Kolmogorov^{Lean}.*
3. *The one-point mixture is the identity^{Lean}.*
4. *Iterated countable mixing is order-independent^{Lean}.*

Restricted to countable-mixture presentations, the data of a monad are all in place: the category of locales, Val restricted

to countable mixtures as the functor, the point valuations as the unit, and countable mixing as the multiplication. Proposition 48 verifies exactly the laws this needs: item 1 keeps the multiplication inside the fragment, item 2 is its associativity, item 3 its unit law, and item 4 its commutativity. What is not yet settled is whether this descends to valuations themselves, since a valuation can admit more than one countable-mixture presentation, and the diffuse part of Section 4 is not captured by any countable presentation at all.

9 Related Work

Informal precedents. Pairing probability with intuitionistic logic is not a new idea, though the literature on it is scattered. Weatherston develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherston 2003]. His axioms match our valuation axioms exactly, with negation derived rather than assumed, so the object itself and the bound $v(a) + v(\neg a) \leq 1$ are his. What is new here is the decomposition of that bound’s slack into dnGap and deMorganGap and its determinacy from v alone (Theorem 6). Paris gives an informal theorem that any modular assignment of non-classical truth values is exactly a Dempster–Shafer belief function [Paris 1994]. We mechanize the analogous fact for our intuitionistic setting: the 2-monotone case follows directly from modularity^{Lean}, and the full tower is Theorem 23. The identification $\text{Bel} = P(\Box \cdot)$ connecting belief functions [Dempster 1967; Shafer 1976] to modal logic is established [Ruspini 1986]. Section 7 shows this identification is not a separate axiom for us but a special case of our frame-level slack, recovered automatically once the frame is a topology and $\neg$ is read as the interior of the complement^{Lean}.

On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers’s work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008]. These normally build in Scott continuity and target additive, Riesz-style goals. Section 4 shows Scott continuity to be a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. Cox, Jaynes, Halpern’s counterexample, and the Van Horn and Arnborg–Sjödín repairs [Arnborg and Sjödín 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003] are classical throughout, and Cox’s and Van Horn’s derivations both lean on double-negation elimination. Section 6 proves the sum-rule half without that move, at the price of positing modularity as an axiom rather than deriving it (Theorem 29). Van Horn observes that dropping axiom R3 yields an interval-valued theory, exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either characterization theorem: complement rule $\iff$ excluded middle (Theorem 10), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 27).

Prior formalizations. Two bodies of mechanized work are adjacent, and neither overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations differ from ours in carrying Scott continuity, and their monad is the Scott-continuous instance of the question examined in Section 8. Our development drops the continuity their base structure builds in, and the two mechanizations do not otherwise overlap. On the domain-theoretic side, commutativity of the full valuations monad on the category of directed-complete partial orders is open. A hierarchy of commutative submonads, generated from simple valuations (finite mixtures of point valuations), has been constructed instead [Jia et al. 2021]. Our countably-presented fragment, the valuations given as countable mixtures (Proposition 48), is the frame-side analogue.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason’s theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Orthomodular lattices give up distributivity itself, while our frames stay distributive by definition (Definition 1), locating all the non-classical behavior in the failure of excluded middle alone. Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Where Walley’s coherent lower previsions posit the belief/plausibility gap directly as a coherence primitive, our slack (Section 2) is derived, forced by modularity and negation rather than assumed. Fritz’s synthetic Markov-category framework [Fritz 2020] varies the ambient structure categorically rather than order-theoretically, axiomatizing conditional independence and sufficient statistics abstractly across discrete, measure-theoretic, and Gaussian probability alike. It remains unmechanized and is orthogonal to our variation in the underlying logic. Wolpert and Kinney model mathematics itself as a stochastic process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: slack-free credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 27 is the exact logical characterization of their agreement, and its mechanization.

10 Conclusion

In this work, we take seriously the Cox–Jaynes reading of probability as *extended logic*, and make the extension a construction with the logic of events as a parameter. The construction fixes an event algebra (the chosen logic in algebraic form) and takes as its assignments of degrees of certainty the valuations: the maps from events to numbers satisfying monotonicity, normalization, and modularity. We specify the two conditions that qualify the result as a probability calculus: when events are certain (0 or 1 certainty), the valuations must reconstruct the underlying event logic, and the value of the construction at classical logic must be the classical theory.

The motivation for developing this extension comes from the study of program behaviour in terms of outcome probabilities. Many properties of programs are only semidecidable. However, the Boolean event algebra underlying Kolmogorov’s axioms imposes the law of excluded middle (and therefore decidability) on the structure of logical connectives applied to events. This makes it not a good fit for reasoning about events that are not decidable, such as, e.g. the termination of the halting program (which, we demonstrate, also becomes decidable when reasoning in classical probability). The extension process we propose, when fixing its logic parameter to be intuitionistic, does not conflict with the underlying event logic of computation by construction, and is capable of reasoning about the possibility of non-termination.

Substituting the Heyting event algebra (and therefore intuitionistic logic) for the Boolean algebra representing the classical logic case changes only the consequences that excluded middle supplied. Both qualifying conditions hold at this parameter value (Sections 2 and 3), and classicality enters only through the complement rule (Theorem 10) and its dynamic counterpart, the agreement of Dempster and Bayesian conditioning (Theorem 27). The complement rule degrades to the inequality $v(a) + v(\neg a) \leq 1$. The deficit in this inequality, for which we re-purpose the term *slack*, is a new quantity: the value of the undecided region, given by the boundary mass in the measure model. Slack is submodular yet does not determine the valuation. The weaker logic also adds the two-gap decomposition of slack, the atomic/diffuse divide, and a belief-function reading through $\neg\neg$.

Everything else holds verbatim on frames: conditioning, the product rule, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself. On finite frames, every valuation is a classical mass function on the frame’s points. The product-rule half of Cox’s theorem is independent of the logic while the sum-rule half must be posited, so the direct transcription of the classical Cox statement is unsatisfiable. Every result is machine-checked in Lean 4 over mathlib.

Toward a monad. Section 8 lays the groundwork for a categorical exploration of the construction: Val is a mechanized functor on locales, and its countably-presented fragment composes as kernels, though whether that composition is canonical on valuations rather than their presentations remains open. A commutative monad structure, where it exists, would be the intuitionistic counterpart of the Giry monad and a step toward compositional semantics for probabilistic programs with semidecidable events. One candidate formulation exists: the space of Scott-continuous valuations is the classifying locale of a geometric theory [Coquand and Spitters 2009; Vickers 2008]. Deleting its continuity axiom still leaves a geometric theory, whose points, read classically, are exactly our valuations. Its multiplication, however, must integrate against a merely modular valuation, and the diffuse part of Example 17 is exactly what such integration fails to capture. We offer the formulation as a target, not a result. Beyond intuitionistic logic, no ambient category sending a logic to its probability calculus functorially is evident from our construction, and this question remains open.

Other open problems and future work. A natural extension of our mechanization effort involves completing the Aczél development, and mechanizing the Dutch-book theorem [Weatherson 2003]. Both require substantial additional effort and were deemed out of scope for this work. To extend our mechanization effort with preliminary monad results, we would quotient the countable presentations (Proposition 48) into a Kleisli category of countably-presented kernels, and mechanize the classifying-locale formulation (which needs classifying locales of geometric theories, infrastructure mathlib currently lacks).

Some related mathematical problems remain open as well. The first is developing a representation of the diffuse part on non-spatial frames. The second is deriving modularity from a criterion more primitive than the mixture characterization (Theorem 16), knowing it cannot come from disjunction data alone (Theorem 29).

The construction method applies beyond intuitionistic logic: state the classical desiderata, delete the axioms the target logic no longer licenses, and check what survives, what splits, and what becomes unsatisfiable. Linear logic is the natural next case: its event algebras are quantales, a valuation there should grade the consumption of evidence, and the analogues of our representation theorems and of slack, for the failure of contraction, remain open [Danos and Ehrhard 2011].

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A Values in the Extended Reals

Valuations take values in the extended non-negative reals $[0, \infty]$, following the measure-theoretic convention of mathlib. As a result, sums indexed by an arbitrary family are unconditionally defined, with no summability hypothesis, and they commute over dependent sums: for a function f on a sigma type $\Sigma a, \beta(a)$, one has $\sum_p f(p) = \sum_a \sum_b f(a, b)$ with no side condition. This is what lets the countable-mixture results of Section 10 hold as stated. Additionally, subtraction is truncated at 0. This never affects the main text, since every subtraction there is applied to a pair already ordered by monotonicity, as with the mass of Section 4, so the truncation is never triggered.

B Lean Statements

B.1 Valuations on Frames

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b =
    toFun (a ⊔ b) + toFun (a ⊓ b)

theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v aᶜ ≤ 1
```

```

1431 theorem Valuation.add_compl_eq_sup (v :
1432   Valuation Ω) (a : Ω) :
1433   v a + v ac = v (a ⊔ ac)
1434
1435 theorem Valuation.le_one (v : Valuation Ω)
1436   (a : Ω) : v a ≤ 1
1437
1438 theorem Valuation.
1439   slack_eq_dnGap_add_deMorganGap (v :
1440     Valuation Ω) (a : Ω) :
1441     v.slack a = v.dnGap a + v.deMorganGap
1442       a
1443
1444 theorem Valuation.slack_eq_sub_add_sub (v
1445   : Valuation Ω) (a : Ω) :
1446   v.slack a = (v acc - v a) + (1 - (v a
1447     cc + v ac))
1448
1449 theorem Valuation.le_compl_compl (v :
1450   Valuation Ω) (a : Ω) : v a ≤ v acc
1451
1452 theorem Valuation.slack_submodular (v :
1453   Valuation Ω) (a b : Ω) :
1454   v.slack (a ⊔ b) + v.slack (a ⊓ b) ≤ v.
1455     slack a + v.slack b
1456
1457 def Order.Frame.IsDeMorgan (Ω : Type*) [
1458   Order.Frame Ω] : Prop :=
1459   ∀ a : Ω, acc ⊔ ac = ⊤
1460
1461 theorem Valuation.
1462   slack_eq_dnGap_of_isDeMorgan
1463   (hΩ : Order.Frame.IsDeMorgan Ω) (v :
1464     Valuation Ω) (a : Ω) :
1465     v.slack a = v.dnGap a
1466
1467 noncomputable def Valuation.condVal (v :
1468   Valuation Ω) (b : Ω) (hb : v b ≠ 0) :
1469   Valuation Ω
1470
1471 theorem Valuation.condVal_mul (v :
1472   Valuation Ω) (b : Ω) (hb : v b ≠ 0) (a
1473   : Ω) :
1474   v.condVal b hb a * v b = v (a ⊓ b)
1475
1476 theorem Valuation.condVal_symm (v :
1477   Valuation Ω) {a b : Ω}
1478   (ha : v a ≠ 0) (hb : v b ≠ 0) :
1479   v.condVal b hb a * v b = v.condVal a
1480     ha b * v a
1481
1482 theorem Valuation.cond_add_compl_le (v :
1483   Valuation Ω) (a b : Ω) :
1484   v (a ⊓ b) + v (ac ⊓ b) ≤ v b
1485

```

```

theorem Valuation.
1486   add_compl_eq_one_of_complemented (v :
1487     Valuation Ω) {a : Ω}
1488     (ha : a ⊔ ac = ⊤) : v a + v ac = 1
1489
1490 variable [CompleteBooleanAlgebra Ω]
1491
1492 theorem classical_additivity (v :
1493   Valuation Ω) (a : Ω) :
1494   v a + v ac = 1
1495
1496 variable [CompleteBooleanAlgebra Ω]
1497
1498 theorem classical_slack_zero (v :
1499   Valuation Ω) (a : Ω) :
1500   v.slack a = 0
1501
1502 noncomputable def Valuation.mix (w : ι → ℝ
1503   ≥ 0∞) (hw : ∑ i, w i = 1)
1504   (v : ι → Valuation Ω) : Valuation Ω
1505

```

B.2 Where Classicality Lives

```

def Valuation.HasClassicalNegation (v :
1509   Valuation Ω) : Prop :=
1510   ∀ a, v a + v ac = 1
1511
1512 theorem hasClassicalNegation_of_em (hem :
1513   ∀ a : Ω, a ⊔ ac = ⊤)
1514   (v : Valuation Ω) : v.
1515     HasClassicalNegation
1516
1517 theorem em_of_forall_hasClassicalNegation
1518   (h : ∀ v : Valuation Ω, v.
1519     HasClassicalNegation) : ∀ a : Ω, a
1520     ⊔ ac = ⊤
1521
1522 theorem sharp_iff_point (v : Valuation Ω)
1523   :
1524   v.IsSharp ↔ ∃ (J : Order.Ideal Ω) (hJ
1525     : J.IsPrime) (htop : ⊤ ∉ J),
1526     v = Ideal.toValuation J hJ htop
1527
1528 theorem
1529   sharp_scott_iff_completelyPrimePoint (v
1530     : Valuation Ω) :
1531   v.IsSharp ∧ v.ScottContinuous ↔
1532   ∃ (J : Order.Ideal Ω) (hJ : Ideal.
1533     IsCompletelyPrime J) (htop : ⊤ ∉
1534     J),
1535   v = Ideal.toValuation J hJ.1 htop
1536

```

B.3 Valuation Representation

1541	noncomputable def Valuation.mass (v :	(hsc : v T = $\bigsqcup$ (S : Finset P), v (S.	1596
1542	Valuation (LowerSet P)) (p : P) : $\mathbb{R}_{\geq 0\infty}$	sup LowerSet.Iic)) : IsPurelyAtomic	1597
1543		v	1598
1544	theorem Valuation.eq_sum_mass (v :		1599
1545	Valuation (LowerSet P)) (U : LowerSet P	theorem tsum_mass_le (v : Valuation (	1600
1546	) :	LowerSet P)) : $\sum' p, v.mass\ p \leq v\ T$	1601
1547	v U = $\sum p \in U.toFinset, v.mass\ p$		1602
1548			1603
1549	theorem Valuation.sum_mass (v : Valuation	B.4 The Dempster–Shafer Reading	1604
1550	(LowerSet P)) : $\sum p, v.mass\ p = 1$	theorem Valuation.inclusion_exclusion (v :	1605
1551		Valuation Ω) (s : Finset ι)	1606
1552	noncomputable def Valuation.toPMF (v :	(a : $\iota \rightarrow \Omega$) : v (s.sup a) + v.iesEven	1607
1553	Valuation (LowerSet P)) : PMF P	s a = v.iesOdd s a	1608
1554			1609
1555	theorem Valuation.exists_pmf_presentation	theorem Valuation.infty_monotone (v :	1610
1556	{ Θ : Type u} [Order.Frame Θ] [Finite Θ]	Valuation Ω) (s : Finset ι) (a : $\iota \rightarrow \Omega$)	1611
1557	(v : Valuation Θ) :	:	1612
1558	$\exists$ (P : Type u) (_ : PartialOrder P) (_	v.iesOdd s a $\leq v$ ((s.sup a) ^{cc}) + v.	1613
1559	: Fintype P) (_ : DecidableEq P)	iesEven s a	1614
1560	(e : $\Theta \approx_o$ LowerSet P),		1615
1561	($\forall a, v\ a = \sum p \in (e\ a).toFinset, (v$	theorem Valuation.two_monotone (v :	1616
1562	.mapEquiv e).mass p) $\wedge$	Valuation Ω) (a b : Ω) :	1617
1563	$\sum p, (v.mapEquiv e).mass\ p = 1$	v a + v b $\leq v$ (a $\sqcup$ b) ^{cc} + v (a $\sqcap$ b)	1618
1564			1619
1565	noncomputable def deltaPoint (p : P) :	theorem Valuation.plausibility_sub_self (v	1620
1566	Valuation (LowerSet P)	: Valuation Ω) (a : Ω) :	1621
1567		v.plausibility a - v a = v.slack a	1622
1568	theorem Valuation.eq_mix_deltaPoint (v :		1623
1569	Valuation (LowerSet P)) :	theorem Valuation.sup_compl_decomp (v :	1624
1570	v = Valuation.mix v.mass v.sum_mass (	Valuation Ω) (a b : Ω) :	1625
1571	fun p => deltaPoint p)	v (a $\sqcup$ b ^c) = v b ^c + (v (a $\sqcap$ b) + v.	1626
1572		emGap a b)	1627
1573	theorem		1628
1574	exists_valuation_not_point_representable	noncomputable def Valuation.dempsterCond (	1629
1575	:	v : Valuation Ω) (b : Ω)	1630
1576	$\exists v : \text{Valuation (LowerSet } \mathbb{N}), \sum' p, v.$	(hb : v b $\neq 0$) : Valuation Ω	1631
1577	mass p $\neq v\ T$		1632
1578		theorem Valuation.dempsterCond_apply (v :	1633
1579	theorem topIndicator_mass (n : $\mathbb{N}$) :	Valuation Ω) (b : Ω) (hb : v b $\neq 0$)	1634
1580	topIndicator.mass n = 0	(a : Ω) : v.dempsterCond b hb a = (v (	1635
1581		a $\sqcup$ b ^c) - v b ^c) / (1 - v b ^c)	1636
1582	theorem topIndicator_not_scott :		1637
1583	($\sqcup$ n, topIndicator (LowerSet.Iic n)) $\neq$	theorem Valuation.	1638
1584	topIndicator T	dempsterCond_eq_condVal_iff_slack (v :	1639
1585		Valuation Ω) {b : Ω }	1640
1586	theorem eq_tsum_mass_of_scott (v :	(hb : v b $\neq 0$) :	1641
1587	Valuation (LowerSet $\mathbb{N}$))	($\forall a, v.dempsterCond\ b\ hb\ a = v.$	1642
1588	(hsc : v T = $\bigsqcup$ n, v (LowerSet.Iic n))	condVal b hb a) $\leftrightarrow$ v.slack b = 0	1643
1589	: v T = $\sum' n, v.mass\ n$		1644
1590			1645
1591	variable [$\forall p : P, \text{Finite (Set.Iic p)}$]	B.5 Cox's Theorem	1646
1592			1647
1593	theorem Valuation.isPurelyAtomic_of_scott	structure CoxModel (Ω : Type*) [Order.	1648
1594	(v : Valuation (LowerSet P))	Frame Ω] where	1649
1595		p1 : $\Omega \rightarrow \Omega \rightarrow \mathbb{R}$	1650

```

1651   mono_left : ∀ c, Monotone fun a => pl a
1652   c
1653   pl_bot : ∀ c, pl ⊥ c = 0
1654   pl_top : ∀ c, pl ⊤ c = 1
1655   F : ℝ → ℝ → ℝ
1656   conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
1657     b ⊓ c)) (pl b c)
1658   F_assoc : ∀ x y z, F (F x y) z = F x (F
1659     y z)
1660   F_strictMono_left : ∀ y, 0 < y →
1661     StrictMono fun x => F x y
1662
1663   theorem isDeMorgan_lowerSetV : Order.Frame
1664     .IsDeMorgan (LowerSet V)
1665
1666   theorem modularity_irreducible :
1667     ∃ (Ω : Type) (⊥ : Order.Frame Ω) (q :
1668       Ω → ℝ≥0∞),
1669     Monotone q ∧ q ⊥ = 0 ∧ q ⊤ = 1 ∧
1670     (∀ x y : Ω, x ⊓ y = ⊥ → q (x ⊔ y) =
1671       q x + q y) ∧
1672     ¬ (∀ x y : Ω, q x + q y = q (x ⊔ y)
1673       + q (x ⊓ y))
1674
1675   theorem no_disjunction_functional :
1676     ¬ ∃ S : ℝ≥0∞ → ℝ≥0∞ → ℝ≥0∞,
1677       ∀ x y : LowerSet V, qV (x ⊔ y) = S
1678         (qV x) (qV y)
1679
1680   theorem constructive_cox (M : CoxModel Ω)
1681     (hmod : ∀ a b : Ω, M.pl a ⊤ + M.pl b ⊤
1682       = M.pl (a ⊔ b) ⊤ + M.pl (a ⊓ b) ⊤)
1683     :
1684     ∃ (v : Valuation Ω) (g : ℝ → ℝ≥0∞),
1685       StrictMonoOn g (Set.Icc 0 1) ∧ g 0 =
1686         0 ∧ g 1 = 1 ∧
1687         ∀ a, v a = g (M.pl a ⊤)
1688
1689   noncomputable def coxModelENNReal :
1690     CoxModel ℝ≥0∞
1691
1692   theorem Scale.exists_mul_generator :
1693     ∃ G : ℝ → ℝ,
1694       (∀ x y, S.u ≤ x → S.u ≤ y → G (S.F x
1695         y) = G x * G y) ∧
1696       (∀ x y, S.u ≤ x → x < y → G x < G y)
1697
1698   theorem Scale.aczelStatement_cone :
1699     ∃ G : ℝ → ℝ,
1700       (∀ x y, S.u ≤ x → S.u ≤ y → G (S.F x
1701         y) = G x * G y) ∧
1702       StrictMonoOn G (Set.Ici S.u)
1703
1704
1705

```

```

theorem exists_diag_eq {F : ℝ → ℝ → ℝ} {a
  b c : ℝ} (hab : a ≤ b)
  (hcont : ContinuousOn (fun x => F x x)
    (Set.Icc a b))
  (hc : c ∈ Set.Icc (F a a) (F b b)) : ∃
    x ∈ Set.Icc a b, F x x = c

structure Scale where
  F : ℝ → ℝ → ℝ
  hdiv : ∀ c, ∃ x, F x x = c
  hassoc : ∀ x y z, F (F x y) z = F x (F y
    z)
  hpos : ∀ c x, x < F x c
  hmono : ∀ x₁ x₂ y₁ y₂ : ℝ, x₁ ≤ x₂ → y₁
    ≤ y₂ → F x₁ y₁ ≤ F x₂ y₂
  hcont : Continuous fun p : ℝ × ℝ => F p
    .1 p.2
  hbot : ∀ x, Filter.Tendsto (fun z => F x
    z) Filter.atBot (nhds x)
  u : ℝ

```

B.6 Two Grounding Models

```

noncomputable def MeasureTheory.Measure.
  toValuationOpens
  (μ : Measure X) [IsProbabilityMeasure
    μ] : Valuation (Opens X)

theorem toValuationOpens_slack_eq_frontier
  (μ : Measure X)
  [IsProbabilityMeasure μ] (U : Opens X)
  :
  μ.toValuationOpens.slack U = μ (
    frontier (U : Set X))

theorem interiorMeasure_add_compl_le (μ :
  Measure X) [IsProbabilityMeasure μ]
  (S : Set X) : μ.interiorMeasure S + μ.
    interiorMeasure Sᶜ ≤ 1

theorem
  isDeMorgan_opens_iff_extremallyDisconnected
  :
  Order.Frame.IsDeMorgan (Opens X) ↔
    ExtremallyDisconnected X

theorem isDeMorgan_opens_iff_stonean [
  CompactSpace X] [T2Space X] :
  Order.Frame.IsDeMorgan (Opens X) ↔
    ExtremallyDisconnected X

inductive ThreePt : Type
  | ptA : ThreePt
  | ptB : ThreePt

```

| ptZ : ThreePt

theorem frontier_subset_ptZ {U : Set
ThreePt} (hU : IsOpen U) :
frontier U $\subseteq$ {ptZ}

theorem exists_ne_valuation_same_slack :
 $\exists v \ v' : \text{Valuation } (\text{Opens ThreePt}), v.$
slack = v'.slack $\wedge v \neq v'$

theorem haltsOpen_compl : haltsOpen^c = ($\perp$
: Opens **Prop**)

theorem slack_eq_boundary {p : $\mathbb{R}_{\geq 0\infty}$ } (hp1
: p ≤ 1) :
(sierpinskiValuation hp1).slack
haltsOpen
= coinMeasure p (frontier {q : **Prop**
| q})

theorem sharpReadout_not_computable (n : $\mathbb{N}$
) :
 $\neg \text{ComputablePred fun } c : \text{Code} \Rightarrow$
sharpReadout n c haltsOpen = 1

theorem isOpen_wordCylinder (w : List Bool
) : IsOpen (wordCylinder w)

theorem cantorMeasure_wordCylinder (w :
List Bool) :
cantorMeasure (wordCylinder w) = (1 /
2) ^ w.length

theorem
disjoint_wordCylinder_of_not_prefix {v
w : List Bool}
(hv : $\neg v <+ w$) (hw : $\neg w <+ v$) :
Disjoint (wordCylinder v) (
wordCylinder w)

theorem cantorMeasure_biUnion_wordCylinder
{F : Finset (List Bool)}
(hF : PrefixFree F) :
cantorMeasure ($\bigcup w \in F, \text{wordCylinder } w$
) = $\sum w \in F, (1 / 2 : \mathbb{R}_{\geq 0\infty}) ^ w.$
length

B.7 The Monad Question

def Valuation.comap (f : FrameHom $\Gamma \ \Omega$) (v
: Valuation Ω) : Valuation Γ

theorem Valuation.comap_id (v : Valuation
 Ω) : v.comap (FrameHom.id Ω) = v

theorem Valuation.comap_comp (f : FrameHom
 $\Gamma \ \Omega$) (g : FrameHom $\Delta \ \Gamma$)
(v : Valuation Ω) : v.comap (f.comp g)
= (v.comap f).comap g

theorem exists_rect_decomposition (W :
LowerSet ($\mathbb{N} \times \mathbb{N}$)) :
 $\exists (N : \mathbb{N}) (A \ B : \mathbb{N} \rightarrow \text{LowerSet } \mathbb{N}),$
W = (Finset.range N).sup fun i => A
i $\times^s$ B i

theorem Valuation.eq_on_finsetSup_of_eq_on
{G : Set Ω }
(hG : $\forall g \in G, \forall g' \in G, g \sqcap g' \in G$) (
u u' : Valuation Ω)
(h : $\forall g \in G, u \ g = u' \ g$) { $\iota : \text{Type}^*$ }
[DecidableEq ι]
(s : Finset ι) (a : $\iota \rightarrow \Omega$) (ha : $\forall i \in$
s, a i $\in G$) :
u (s.sup a) = u' (s.sup a)

theorem Valuation.product_unique (v w :
Valuation (LowerSet $\mathbb{N}$))
{u u' : Valuation (LowerSet ($\mathbb{N} \times \mathbb{N}$))}
(hu : $\forall A \ B : \text{LowerSet } \mathbb{N}, u (A \times^s B) =$
v A * w B)
(hu' : $\forall A \ B : \text{LowerSet } \mathbb{N}, u' (A \times^s B)$
= v A * w B) : u = u'

noncomputable def ultrafilterValuation (W
: Ultrafilter ι) : Valuation (Set ι)

theorem rect_mem_iff { $\alpha \ \beta : \text{Type}^*$ } {W :
Ultrafilter ($\alpha \times \beta$)}
{S : Set α } {T : Set β } :
S $\times^s$ T $\in W \leftrightarrow S \in W.$ map Prod.fst $\wedge T \in$
W.map Prod.snd

theorem product_valuation_not_unique :
 $\exists u \ u' : \text{Valuation } (\text{Set } (\mathbb{N} \times \mathbb{N})),$
($\forall S \ T : \text{Set } \mathbb{N}, u (S \times^s T) = u' (S \times$
 $^s T)) \wedge u \neq u'$

theorem fubini_fails_for_valuations :
 $\exists (v \ w : \text{Valuation } (\text{Set } \mathbb{N})) (u \ u' : \text{Valuation } (\text{Set } (\mathbb{N} \times \mathbb{N}))),$
($\forall S \ T : \text{Set } \mathbb{N}, u (S \times^s T) = v \ S * w$
T) $\wedge$
($\forall S \ T : \text{Set } \mathbb{N}, u' (S \times^s T) = v \ S *$
w T) $\wedge u \neq u'$

noncomputable def Valuation.mixCountable {
 $\iota : \text{Type}^*$ } (w : $\iota \rightarrow \mathbb{R}_{\geq 0\infty}$)


```

1871 (hw :  $\sum' i, w\ i = 1$ ) (v :  $\iota \rightarrow$ 
1872   Valuation  $\Omega$ ) : Valuation  $\Omega$ 
1873
1874 theorem Valuation.
1875   mixCountable_mixCountable { $\iota$  : Type*} {
1876      $\kappa$  :  $\iota \rightarrow$  Type*}
1877   (w :  $\iota \rightarrow \mathbb{R}_{\geq 0\infty}$ ) (hw :  $\sum' i, w\ i = 1$ )
1878   (u :  $\forall i, \kappa\ i \rightarrow \mathbb{R}_{\geq 0\infty}$ ) (hu :  $\forall i, \sum' j,$ 
1879     u i j = 1)
1880   (v :  $\forall i, \kappa\ i \rightarrow$  Valuation  $\Omega$ ) :
1881   Valuation.mixCountable w hw
1882     (fun i => Valuation.mixCountable (
1883       u i) (hu i) (v i))
1884   = Valuation.mixCountable (fun p :  $\Sigma$ 
1885     i,  $\kappa\ i \Rightarrow w\ p.1 * u\ p.1\ p.2$ )
1886     (tsum_sigma_mul_eq_one w hw u hu
1887       ) (fun p => v p.1 p.2)
1888
1889 theorem Valuation.mixCountable_unique (v :
1890   Valuation  $\Omega$ ) :
1891   Valuation.mixCountable (fun _ : PUnit
1892     => 1) (by simp) (fun _ => v) = v
1893
1894 theorem Valuation.mixCountable_comm { $\iota$   $\kappa$  :
1895   Type*}
1896   (w :  $\iota \rightarrow \mathbb{R}_{\geq 0\infty}$ ) (hw :  $\sum' i, w\ i = 1$ ) (u
1897     :  $\kappa \rightarrow \mathbb{R}_{\geq 0\infty}$ ) (hu :  $\sum' j, u\ j = 1$ )
1898   (v :  $\iota \rightarrow \kappa \rightarrow$  Valuation  $\Omega$ ) :
1899   Valuation.mixCountable w hw (fun i =>
1900     Valuation.mixCountable u hu (v i))
1901   = Valuation.mixCountable u hu
1902     (fun j => Valuation.mixCountable
1903       w hw (fun i => v i j))
1904
1905
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```

C The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 30. The target shape is Aczél's associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ regruates to multiplication, $G(F\ x\ y) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G . `mathlib` has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A Scale bundles: associativity of F , *divisibility* $\forall c \exists x, F\ x\ x = c$, positivity on the cone $x < F\ x\ c$, joint order-preservation, joint continuity, a unit u , and identity at $-\infty$, meaning $F\ x\ z \rightarrow x$ as $z \rightarrow -\infty$. Order-preservation is required by the additivity proof below. Divisibility yields the 2^n -th roots used in the dyadic construction below. Without it the generator is not definable. It is assumed directly in the bundle, though its interval form follows from the intermediate value theorem (`exists_diag_eq`). Non-vacuity:

log-sum-exp, $F\ x\ y = \log(e^x + e^y)$ with $u = 0$, satisfies every field of the bundle (`logSumExpScale`), with intended generator globally `exp`.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (`Fpow`). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$. Commutativity is not assumed, and follows instead as a corollary of the generator. Unboundedness of the iterates is derived: were they bounded, monotone convergence would give a fixed point $F\ L\ a = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility yields a tower of 2^n -th roots $u \succ u^{1/2} \succ u^{1/4} \succ \dots$ (`droot`), where 2^ℓ copies of the finer root compose under F to one copy of the coarser root. Counting the number of copies of the n -th root not exceeding b (`dcount`, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \text{dcount}(b, n)/2^n$. The refinement inequality $2N + 1 \leq N' \leq 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n + 1$, makes (g_n) monotone and Cauchy, and the generator is its limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all sorry-free, standard axioms). On the cone $\{x \mid u \leq x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(F\ x\ y) = g(x) + g(y)$, since $\text{dcount}\ x + \text{dcount}\ y + 1 \leq \text{dcount}(F\ x\ y) \leq \text{dcount}\ x + \text{dcount}\ y + 3$ and the 2^{-n} error term vanishes. And strict monotonicity, proved without any continuity of g : a root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(F\ x\ y) = G(x) G(y)$ on the cone (`exists_mul_generator`), the conclusion of Aczél's theorem restricted to the cone (`aczelStatement_cone`). The reorientation $\hat{G} = \exp(-g)$ gives the $[0, 1]$ -valued form used in Section 6.

Remaining gap. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim needs the off-cone extension (a group completion) and continuity of the extension. This is open. It is not vacuous: the `log-sum-exp` example admits a global continuous generator (`hasOrderedGenerator_logSumExp`). Of Hölder's classical hypotheses, the Archimedean property is proved, and is what enables the embedding into $\mathbb{R}$. The interval form of divisibility follows from the intermediate value theorem. Only the embedding's global regularity remains.